Colter Snyder

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Hiring Team
Google

Re: Software Engineer III, Infrastructure, Platforms Infrastructure Engineering

Dear Hiring Team,

The infrastructure behind Google's AI and compute platforms spans everything from service code to TPUs, and that full stack is exactly the range I enjoy working across. I am a software engineer at Innovative Scientific Solutions, where I build back-end calculation services for simulation platforms used by defense agencies. I hold an M.S. in Computer Science from the Colorado School of Mines, and I would like to bring my experience with services, performance, and hardware to Platforms Infrastructure Engineering.

At ISSI I build Rust and Python services that communicate over WebSockets, store data in MongoDB, and deploy in Docker containers on Linux. When the team needed to understand how those services performed, I spearheaded an internal benchmarking tool in Python, first as a command-line application and then with a GUI front end. It runs any combination of our 13 effect calculations and charts the results, and it exposed a tenfold slowdown in our interface to an external library, with graphs clear enough to take straight to leadership. It also saved what would likely have been weeks of manual testing with hand-written JSON inputs. Beyond performance work, I port legacy physics models from Fortran and Python to Rust while preserving their numerical behavior, and I lead development of individual features from design documents through automated tests. I review teammates' merge requests regularly, since every change requires a separate reviewer and merger.

My interest in infrastructure goes below the operating system. In my own time I design FPGA logic in Verilog, program in C, and write 6502 and Z80 assembly, which has given me a working sense of how software meets compute hardware. I have also co-taught Data Structures and Algorithms in C++ to 120 students, and diagnosing a 2,000-test suite during my Tyler Technologies internship, where I restored about 1,500 tests to passing, taught me to track problems to their source rather than their symptoms.

I am eager to grow into larger-scale distributed systems, and I can think of no better place to do it than the team building Google's computing foundation. I live in Lakewood and would be glad to work from Google's Boulder office or remotely. Thank you for your time and consideration.

Sincerely,

Colter Snyder